
Mathematics in the Modern World

Lecture Notes

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Preface

I originally prepared these lecture notes for a class in *Mathematics in the Modern World* composed entirely of biology majors. I have therefore designed the discussion, examples, and exercises specifically with that audience in mind. The aim is not merely to help students complete a required mathematics course, but to give them ideas and habits of reasoning that will remain useful as they progress through university and in their later scientific, professional, and everyday lives. Where practicable, the notes connect mathematics with biology, health, data, technology, and the kinds of decisions that future biologists and healthcare professionals may encounter.

The notes have also been designed to comply fully with the curriculum guidelines for *Mathematics in the Modern World* issued by the Philippine Commission on Higher Education (CHED), including the prescribed learning outcomes and subject coverage. Within that framework, I have tailored the explanations, applications, and supplementary examples to the needs and interests of biology students. This alignment does not imply that CHED has reviewed or endorsed these independently prepared notes.

These notes are a work in progress. Some chapters may be incomplete or absent, and existing material may be revised as the course develops. Reports of errors, corrections, and suggestions for improvement are welcome. Please send them to

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1 Mathematics in Our World

Mathematical ability is often equated with numerical fluency. The association is understandable: much of school mathematics concerns arithmetic, algebra, trigonometry, statistics, probability, and calculus. Numerical skill matters, but mathematics extends beyond calculation. It studies patterns, structures, quantities, shapes, change, relationships, and uncertainty. It also supplies a language for describing these subjects and a logical framework for reasoning about them.

Learning Outcomes

After completing this chapter, you should be able to:

- identify examples of patterns and regularities in nature and everyday life;
- explain why mathematics is more than numerical calculation;
- distinguish a mathematical model from the reality it represents;
- recognise examples of inductive and deductive reasoning;
- describe some practical, intellectual, and aesthetic dimensions of mathematics; and
- explain how mathematics contributes to biology and modern technology.

Planning an ordinary morning. Some uses of mathematics in everyday life are fairly obvious. Suppose your first class begins at 8:00 a.m. What time should you wake up to arrive on time? You might estimate how long it takes to get out of bed, take a shower, get dressed, prepare your belongings, eat breakfast, and travel to school. At first, this appears to be a simple arithmetic problem: add the estimated duration of each activity, then subtract the total from the class's starting time.

But what you are doing is not merely arithmetic. You are also using probability, estimation, and decision-making under uncertainty. For example, if you take public transportation, your travel time depends partly on how frequently buses, jeepneys, or tricycles pass your stop. If vehicles arrive regularly, you may expect only a short waiting time. A long wait is possible, but you may consider it unlikely. Your estimate may also change depending on whether your class is held on a weekday or a Saturday, since traffic conditions are usually different.

You may also consider the possibility of rain, road construction, long queues, or unexpected delays. You might then add an extra fifteen or twenty minutes as a safety allowance. How large that allowance should be depends on how uncertain the journey is and how strongly you wish to avoid being late.

You have constructed a simple **mathematical model** of your morning. The model does not reproduce every detail of reality. Instead, it identifies the factors that matter most and uses them to guide a decision.

We sometimes avoid making these calculations ourselves by consulting navigation applications such as Google Maps or Waze. These applications work precisely because they combine several areas of mathematics:

- **Geometry and coordinate systems** represent locations, distances, and directions.
- **Graph theory** represents intersections as vertices and roads as weighted edges.
- **Optimisation algorithms**, including Dijkstra's algorithm [1] and A* [2], search for efficient routes through the resulting network.
- **Statistics and calculus** help estimate travel times and account for changing traffic conditions.

The estimated journey time is therefore produced by a mathematical model of location, networks, uncertainty, and change—not by arithmetic alone.



Figure 1: A conceptual illustration of shortest-path search across an urban road network represented as a weighted graph. The green and red markers denote the origin and destination, the white line shows the selected route, and the coloured regions indicate parts of the network explored during the search. Dijkstra's algorithm is a foundational graph-theoretic method for solving this kind of problem, although modern navigation applications employ more sophisticated variants.

Task

Watch the YouTube video [Why Navigation Apps Always Find the Perfect Route?](#) [3].

You will not be expected to implement Dijkstra's algorithm or A* as shown in the video. The example simply illustrates how familiar applications use mathematics behind the scenes. Real navigation systems extend these basic ideas to account for live traffic, road restrictions, and other practical conditions.

1.1 Patterns in Nature

Mathematics does more than help us make practical decisions. It also helps us recognise, describe, and investigate patterns—including patterns that people find beautiful.

Task

Watch the YouTube video [Nature by Numbers](#) [4].

1.2 The Golden Ratio

The video above brings together the famous Fibonacci sequence, the golden ratio, and several patterns associated with them. These ideas are closely connected, and the golden ratio provides the central thread linking much of the discussion.

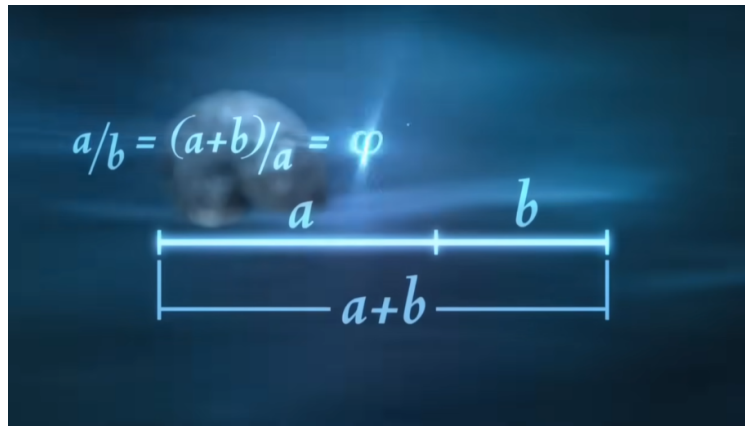
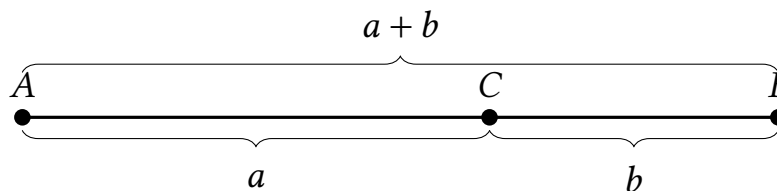


Figure 2: A frame from [Nature by Numbers](#) illustrating the golden ratio [4].

In Euclid's *Elements* [5], a straight line is said to be divided in the *golden ratio*¹ when

The whole is to the greater segment as the greater segment is to the lesser.



Let's unpack this definition. Suppose a line segment AB is divided at a point C . Let the greater segment AC have length a and the lesser segment CB have length b . The whole segment AB then has length $a + b$.

¹Euclid calls this the *extreme and mean ratio*; here we use the modern term *golden ratio*.

Euclid's definition states that the ratios $a+b/a$ and a/b are equal. Their common value is the golden ratio, usually denoted by the Greek letter φ . Thus,

$$\varphi = \frac{a+b}{a} = \frac{a}{b}.$$

This relationship allows us to calculate the numerical value of φ :

$$\varphi = \frac{a+b}{a} = 1 + \frac{b}{a} = 1 + \frac{1}{\varphi}.$$

Multiplying by φ gives

$$\varphi^2 - \varphi - 1 = 0.$$

The quadratic formula produces two roots. Because a ratio of lengths is positive, we retain the positive root:

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339887.$$

1.3 The Fibonacci Sequence

Fibonacci numbers arise in mathematical descriptions of several natural patterns, particularly some arrangements of leaves, petals, and seeds. Their appearance is not universal; when these patterns do occur, they can emerge from local rules governing growth and spatial arrangement.



Figure 3: A frame from [Nature by Numbers](#) illustrating the Fibonacci sequence [4].

Spiral shells, such as those of the [nautilus](#), are often used to illustrate this idea [6]. Their shape may resemble a logarithmic spiral, but visual resemblance alone does not establish that a shell follows the Fibonacci sequence or the golden ratio exactly.

The golden ratio and the Fibonacci sequence are closely related. Before examining that relationship, let us define the sequence formally.

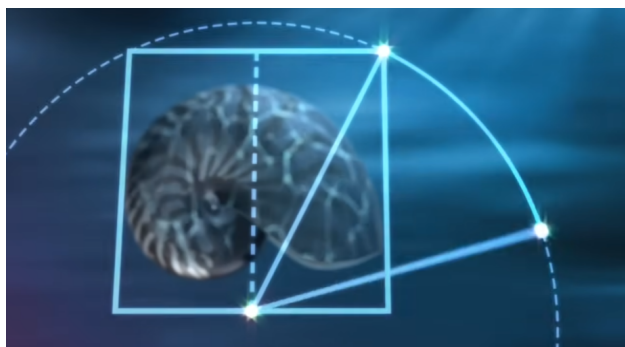


Figure 4: A frame from [Nature by Numbers](#) comparing the spiral of a nautilus shell with a mathematical spiral [4].

Definition: The Fibonacci Sequence

The *Fibonacci sequence* is the sequence $F_0, F_1, F_2, F_3, \dots$ defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

The first few terms are:

$$F_0 = 0 \qquad F_4 = F_3 + F_2 = 2 + 1 = 3,$$

$$F_1 = 1 \qquad F_5 = F_4 + F_3 = 3 + 2 = 5,$$

$$F_2 = F_1 + F_0 = 1 + 0 = 1 \qquad F_6 = F_5 + F_4 = 5 + 3 = 8,$$

$$F_3 = F_2 + F_1 = 1 + 1 = 2 \qquad F_7 = F_6 + F_5 = 8 + 5 = 13.$$

Continuing this process, the sequence begins

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$$

Each term after the first two is the sum of the two preceding terms. Now consider ratios of consecutive nonzero terms:

$$\frac{F_3}{F_2} = \frac{2}{1} = 2, \quad \frac{F_4}{F_3} = \frac{3}{2} = 1.5, \quad \frac{F_5}{F_4} = \frac{5}{3} \approx 1.667,$$

$$\frac{F_6}{F_5} = \frac{8}{5} = 1.6, \quad \frac{F_7}{F_6} = \frac{13}{8} = 1.625, \quad \frac{F_8}{F_7} = \frac{21}{13} \approx 1.615.$$

The ratios alternate above and below 1.618, drawing progressively closer to the golden ratio. This suggests a natural question: do the ratios actually approach φ ?

Fibonacci and the Golden Ratio (Non-examinable)

Recall that

$$F_n = F_{n-1} + F_{n-2}.$$

We are interested in the behaviour of F_n/F_{n-1} as n becomes large. Assume that this ratio approaches a positive limit L :

$$L = \lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}}.$$

Dividing the recurrence relation by F_{n-1} gives

$$\frac{F_n}{F_{n-1}} = 1 + \frac{F_{n-2}}{F_{n-1}} = 1 + \frac{1}{F_{n-1}/F_{n-2}}.$$

The shifted ratio F_{n-1}/F_{n-2} has the same limit L . Taking limits therefore gives

$$L = 1 + \frac{1}{L}.$$

Because $L > 0$, solving $L^2 - L - 1 = 0$ yields

$$L = \frac{1 + \sqrt{5}}{2} = \varphi.$$

A complete proof must also establish that the limit exists; that additional step is beyond the scope of this course.

Inductive and Deductive Reasoning

The Fibonacci example illustrates two complementary forms of mathematical reasoning. By calculating several consecutive ratios and observing that they move closer to 1.618, we use **inductive reasoning** to form the conjecture that the ratios approach the golden ratio. This is persuasive evidence, but checking finitely many terms cannot prove a statement about the entire infinite sequence.

We then use **deductive reasoning**: starting from the Fibonacci recurrence and the assumption that the ratio has a positive limit, we apply algebra and the laws of limits to show that this limit must satisfy $L^2 - L - 1 = 0$, and therefore that $L = \varphi$. A fully rigorous proof must also justify the assumption that the limit exists.

Alternative Methods

The argument above is only one route to the result

$$\lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}} = \varphi = \frac{1 + \sqrt{5}}{2}.$$

Other methods, including techniques from linear algebra, lead to the same conclusion. In assessments, a correct and clearly explained method can receive full credit even if it differs from the method presented in class. You are not expected to discover more advanced approaches, but you are encouraged to think creatively and justify your reasoning.

1.4 The Golden Angle and a Word of Caution

In botany, **phyllotaxis** describes how leaves, seeds, or other plant organs are arranged. In some spiral arrangements, successive organs form an angle close to the **golden angle**

$$\frac{360^\circ}{\varphi^2} \approx 137.5^\circ.$$

This spacing is associated with the familiar Fibonacci-like spirals seen in some flower heads and other plants.



Figure 5: A frame from [Nature by Numbers](#) illustrating a golden-angle pattern in a sunflower head [4].

This pattern is a genuine subject of research in mathematical biology [7], but it should not be treated as a universal rule. Not every plant follows a golden-angle pattern, and plants do not calculate φ . Rather, the pattern can emerge from biological growth processes and interactions among developing structures in limited space.

A word of caution. The golden ratio appears in some natural patterns, but claims about it are often exaggerated. Faces, paintings, buildings, shells, and galaxies are frequently described as “golden” without sufficient evidence.



Figure 6: The Parthenon in Greece [8]. Although the building is often said to have been designed around the golden ratio, historical and geometrical evidence does not support that claim [9–11].

A shape that looks approximately golden is not evidence that the ratio was deliberately used or that it caused the shape to form. A mathematical claim should be supported by clear definitions, reproducible measurements, and—when biology is involved—a plausible biological mechanism.

1.5 Mathematics and Biomedical Imaging

The discussion thus far has shown how mathematical patterns appear in nature. However, one might reasonably argue that the golden ratio is more of an intellectual curiosity than a practical tool. Mathematics, however, also plays a direct and increasingly important role in fields such as medicine.



Figure 7: An illustration of normal vision (left) and vision affected by advanced diabetic retinopathy (right). Early DR may cause no noticeable symptoms.

Figure 7 illustrates what a person might see with a healthy eye and with an eye affected by **diabetic retinopathy** (DR), a complication of diabetes. Diabetes can damage the small blood

vessels in the retina, the light-sensitive layer at the back of the eye. In advanced cases, these vessels may leak or bleed, causing blurred vision, dark floating spots, or even blindness. To screen for the disease, clinicians examine the retina using **fundus photographs**—colour photographs of the back of the eye—such as those shown in Figure 8.

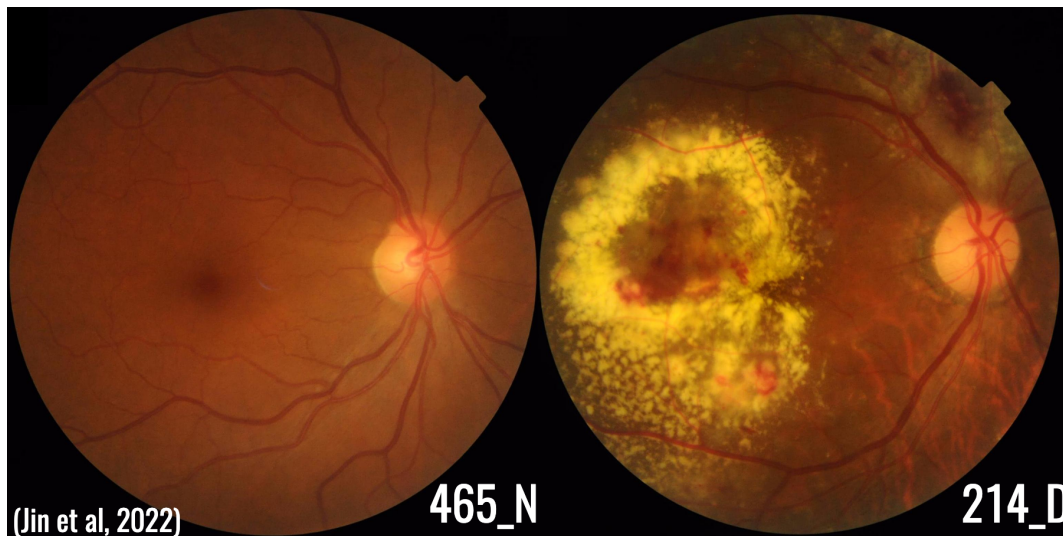


Figure 8: Fundus photographs from the FIVES dataset: a normal retina (left) and a retina affected by diabetic retinopathy (right) [12].

In this example, the image on the right appears much more irregular than the one on the left. It contains large yellow regions and other visible abnormalities. This particular case is relatively easy to distinguish by inspection, but not every case is so obvious. The signs of early disease can be subtle, and a diagnosis must always be made by a qualified clinician.

Many of you may aspire to become doctors or other healthcare professionals. You may also be aware of the shortage of medical professionals throughout the Philippines. When large numbers of patients require screening, computers can help organise their images and identify cases that may need closer attention. Such systems do not replace clinicians; instead, they help clinicians process large amounts of information, care for more patients, and, ultimately, save more lives.

At a glance, one might informally describe the diseased image in Figure 8 as looking “messier.” A computer, however, does not understand such a description. It requires features that can be measured numerically.

One approach is **topological data analysis** (TDA), a mathematical framework often combined with artificial intelligence and computer vision. In my bachelor’s thesis [13], I drew on ideas from several mathematical fields, including linear algebra, graph theory, algebraic topology, and category theory, to convert fundus images into numerical signatures. These signatures can then be supplied to a machine-learning system that classifies an image as either normal or affected by diabetic retinopathy. Figure 9 summarises this process.

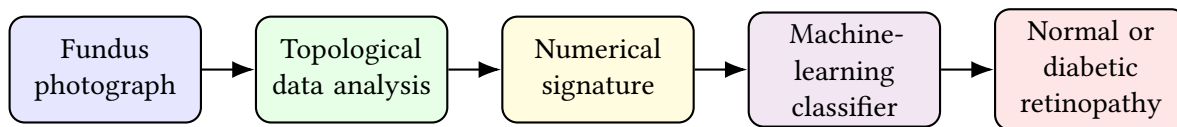


Figure 9: A simplified overview of the TDA-assisted retinal-image classification workflow [13].

This may sound overwhelming, but the central idea is straightforward: TDA can identify simple structural features in an image, such as separate components and loops. It can then summarise these features as a short list of numbers, sometimes called a **topological signature**.

Once each image has a numerical signature, the signatures of different groups can be compared statistically. They can also be supplied to a machine-learning model, such as an artificial neural network.

This does not mean that TDA can diagnose diabetic retinopathy on its own. Digital images are already composed of numerical pixel values, and many computational methods analyse those values directly. TDA provides an additional description that focuses on shape and connectedness. Researchers must therefore test whether this additional information makes a computer-assisted screening system more accurate, robust, or reliable.

Machine Learning and Modern Artificial Intelligence. Machine learning provides much of the foundation for modern artificial intelligence, and many widely used machine-learning systems employ **artificial neural networks**. Despite their biologically inspired name, these are mathematical and computational models rather than literal replicas of a brain.

Building and training them brings several disciplines together. Computer science supplies the algorithms and software; linear algebra represents data and the connections within a network; calculus helps adjust those connections during training; and probability and statistics help describe uncertainty, detect patterns, and evaluate performance. Modern AI therefore rests not on a single technique, but on the interaction of computation and several branches of mathematics.

Key Takeaway

Earlier, we said that mathematics studies patterns, structures, quantities, shapes, change, relationships, and uncertainty. The system described in this section brings all of these ideas together. It searches for patterns in retinal images, represents shapes and structures using numbers, studies how those features change under different conditions, and examines their relationship with disease. Statistics and machine learning then help manage uncertainty when classifying an image.

This is an example of mathematics turning an abstract description of shape into a practical tool that can support medical screening and patient care.

1.6 Is Mathematics an Exact Science?

Mathematics is often described as exact, but that statement needs an important qualification. **Pure mathematics can be exact within its definitions and assumptions.** Once we agree on the rules, a valid proof establishes its conclusion with logical certainty. The real world, however, does not arrive with perfectly defined objects, flawless measurements, or assumptions that are always true.

Reporting a Person's Height

Suppose someone says that he is 5 feet 6 inches tall. This normally does not mean that his height is *exactly* 66 inches. The measurement may have been rounded to the nearest inch; it may also vary slightly with posture, footwear, the measuring instrument, and even the time of day. Nevertheless, “5 feet 6 inches” is usually precise enough for an ordinary conversation or a form asking for someone’s height. Greater precision would add detail, but it would not necessarily be useful for the purpose at hand.

The same principle applies when mathematics is used to study traffic, populations, disease, climate, finance, or biological systems. A **mathematical model** keeps the features judged relevant to a question and leaves others out. It is therefore a simplified representation of reality, not reality itself. Its conclusions depend both on sound mathematics and on how well its assumptions and data represent the situation being studied.

This does not make applied mathematics unreliable. It means that a model should be judged in context: What is it intended to do? How accurate must it be for that purpose? What assumptions does it make? How uncertain are its measurements and predictions? A subway map is not geographically exact, for example, but it can still be an excellent model for finding the correct train.

Exact reasoning, approximate descriptions. Mathematics allows us to reason precisely about a chosen model. It does not guarantee that the model is a perfect copy of the world. A useful model is often not the one containing every possible detail, but the simplest one that is sufficiently accurate for its intended purpose.

1.7 Mathematics and Collaboration

Mathematicians are skilled at working with quantities, detecting patterns, identifying structures, and reasoning from carefully stated assumptions. These abilities are powerful, but they are not sufficient on their own. As the familiar saying warns, “If all you have is a hammer, everything

looks like a nail.” A mathematician who approaches a biological problem using mathematics alone may construct an elegant model while overlooking something biologically important.

Applying mathematics effectively therefore usually requires collaboration with **domain experts**—people with specialised knowledge of the subject being investigated. In biology, these experts may include laboratory researchers, field biologists, clinicians, epidemiologists, and public-health specialists. They help determine which questions matter, which assumptions are reasonable, what data can be collected, and whether a mathematical conclusion makes biological sense.

Mathematicians and statisticians can translate these questions into models, analyse the resulting data, and quantify uncertainty. Computer scientists can develop the algorithms and systems needed to process large or complex datasets. Together, domain expertise, mathematics, and computer science form a common foundation for **data science**, as illustrated in Figure 10. Data science draws on all three areas to extract meaningful information from data and support better decisions [14].

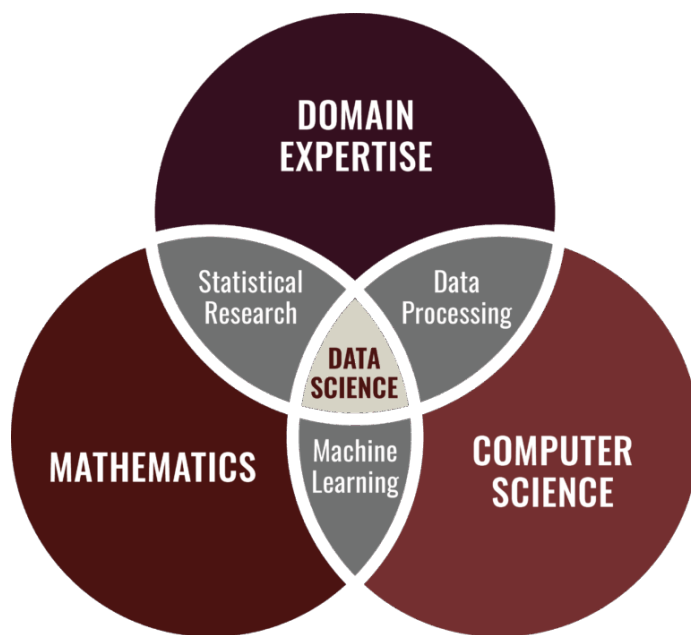


Figure 10: A common view of data science as the intersection of domain expertise, mathematics, and computer science. The overlapping areas also support statistical research, data processing, and machine learning [14].

Many of you may never study advanced mathematics, and you do not need to become mathematicians. Nevertheless, a working understanding of mathematical ideas will help you communicate and collaborate with those who use them. It may also help you recognise when a biological question can be measured, modelled, or investigated in a new way.

Remark

My academic background is primarily in mathematics and computer science. To address the biological aspects of my thesis, I relied on well-established findings from biologists. This experience illustrates why many of today's important discoveries are collaborative efforts involving domain experts, mathematicians, and computer scientists. Although developing expertise in one field remains important, interdisciplinary work becomes much easier when each participant also has some familiarity with the knowledge, methods, and language of the other disciplines.

Some biologists specialise in mathematical or computational work, whereas others focus on laboratory, clinical, or field research. Modern science needs all of them. Important advances often arise when people with different forms of expertise understand one another well enough to work on the same problem.

One day, such a collaboration may help some of you develop a better cancer treatment, improve the management of HIV, or respond to a disease that has not yet emerged. The purpose of learning mathematics here is not merely to pass an examination. It is also to give you another language with which to participate in interdisciplinary science.

1.8 What It Means to Think Mathematically

Being mathematically capable requires more than performing calculations. It includes the ability to:

- recognise patterns and relationships;
- identify relevant information;
- construct and evaluate models;
- reason logically from assumptions;
- account for uncertainty;
- compare possible solutions; and
- explain why a conclusion should be accepted.

Numbers remain indispensable, but they are only part of the story. Mathematics is not simply a collection of techniques for obtaining numerical answers. It is a way of organising information, examining relationships, solving problems, and thinking carefully about the world.

1.9 Check Your Understanding

These questions ask you to use the ideas in the chapter, not merely read about them.

1. **Model an ordinary journey.** List the factors that could affect how long it takes you to travel to class. Which factors would you include in a useful model, and which details could you reasonably leave out?
2. **Investigate a pattern.** Calculate the next three Fibonacci numbers and the ratios of successive terms. Do the new ratios support the conjecture that they approach ϕ ? Explain why this is evidence rather than a complete proof.
3. **Describe an image mathematically.** Compare the two retinal photographs above. Suggest one visible feature that could be measured numerically. Explain what information your proposed measurement captures and what it might overlook.
4. **Question a claim.** Find an everyday or biological claim involving a mathematical pattern. What definition, measurements, or evidence would you require before accepting it?
5. **Defend a broader view of mathematics.** A classmate says that mathematics is merely a collection of formulas for obtaining numerical answers. Write a short response challenging this view. Use two examples from the chapter to explain how mathematics can be expressed, represented, and used, and identify at least one practical, intellectual, or aesthetic dimension of the subject.
6. **Apply mathematics responsibly.** Suppose the developers of a computer-assisted retinal-screening system know that its performance deteriorates when photographs are blurry, but advertise only its overall accuracy. What information should they disclose? Explain how honesty, integrity, and awareness of a model's limitations should affect the use of such a system in healthcare.

Solutions

These are model answers. Other well-justified answers are possible.

1. A useful model might include distance, transport, traffic, weather, waiting time, and road closures. The vehicle's colour could be omitted because it does not materially affect journey time.
2. The next three terms are 34, 55, and 89. The corresponding ratios are

$$\frac{34}{21} \approx 1.6190, \quad \frac{55}{34} \approx 1.6176, \quad \frac{89}{55} \approx 1.6182.$$

These support the conjecture that the ratios approach φ , but finitely many examples cannot prove a statement about the entire sequence.

3. One possible measurement is the number or area of visible yellow lesions. It captures one sign of diabetic retinopathy but may overlook vessel damage, image quality, lesion type, or location.
4. For a claim that a shell follows the golden ratio, we would need defined measurements, a reliable procedure, a suitable sample, and a stated tolerance for closeness to φ . Visual resemblance alone is not sufficient evidence.
5. Mathematics extends beyond calculation. A navigation application represents roads as a network and uses several mathematical tools to select a route; the Fibonacci and golden-ratio examples use numbers, diagrams, and logical relationships to investigate natural patterns. These illustrate practical, intellectual, and aesthetic dimensions of mathematics.
6. The developers should disclose the test data, error rates, and loss of performance on blurry images. They should state that the system supports screening rather than diagnosis and that uncertain cases require clinical review. Integrity requires reporting assumptions, uncertainty, and known limitations—not only a favourable overall accuracy.

Closing Reflection

What decisions have you already made today that involved mathematics, even though you did not consciously perform a written calculation?

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